



AI -ASSISTED LUNAR TERRAIN ANALYSIS & SAFE ROVER NAVIGATION



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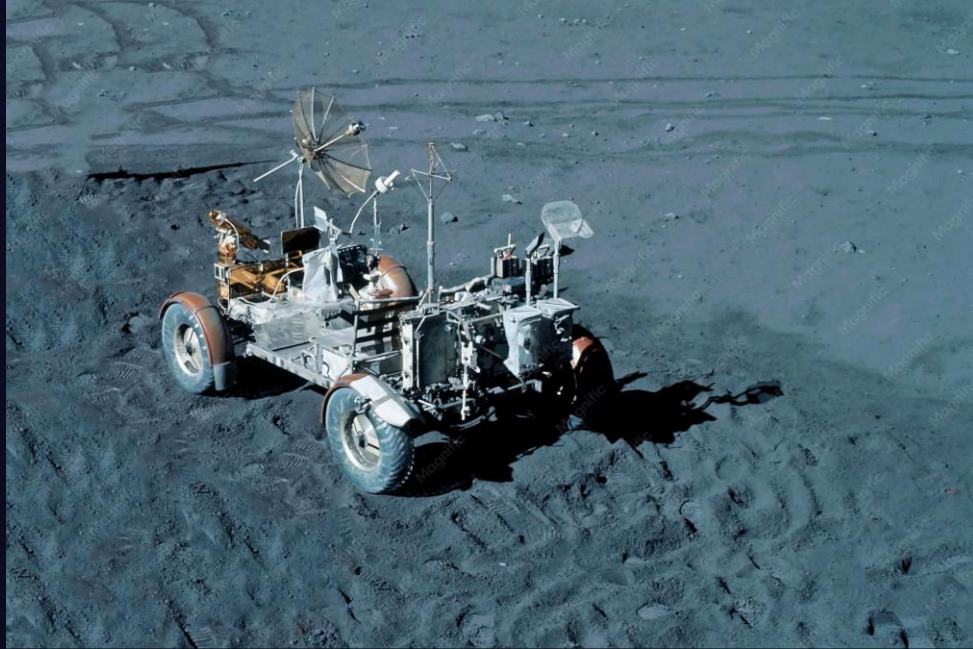
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Lunar AI: Intelligent Landing Site Selection & Rover Navigation

An AI-powered system that analyzes lunar satellite imagery, detects terrain hazards, recommends the safest landing zones, and plans optimal rover paths – built for the next era of Moon and Mars exploration.

Why Landing on the Moon Is Harder Than It Looks



❖ Terrain Hazards

Craters, boulders, steep slopes, and regolith make manual landing site selection extremely risky. A single misjudgment can destroy a mission.

❖ Manual Analysis Is Too Slow

Human experts analyzing satellite imagery take hours or days — far too slow for autonomous or time-critical missions.

- ❑ An AI-assisted, real-time system is essential for safe, scalable lunar exploration.

The Moon Is Back — And the World Is Racing



□ Why This Matters Now

NASA's Artemis Program and ISRO's Chandrayaan missions have reignited global lunar exploration. Safe, autonomous landing and navigation are no longer optional — they are mission-critical requirements.

- Artemis III targets crewed south pole landing by 2026
- ISRO planning Chandrayaan-4 sample return mission
- Autonomous navigation reduces communication delay risks

What the System Must Do



Analyze Lunar Images

Process high-resolution satellite imagery of the lunar surface



Detect Craters & Rocks

Identify surface hazards including craters, boulders, and debris



Terrain Classification

Classify slopes, flat zones, and dangerous terrain types



Recommend Landing Zone

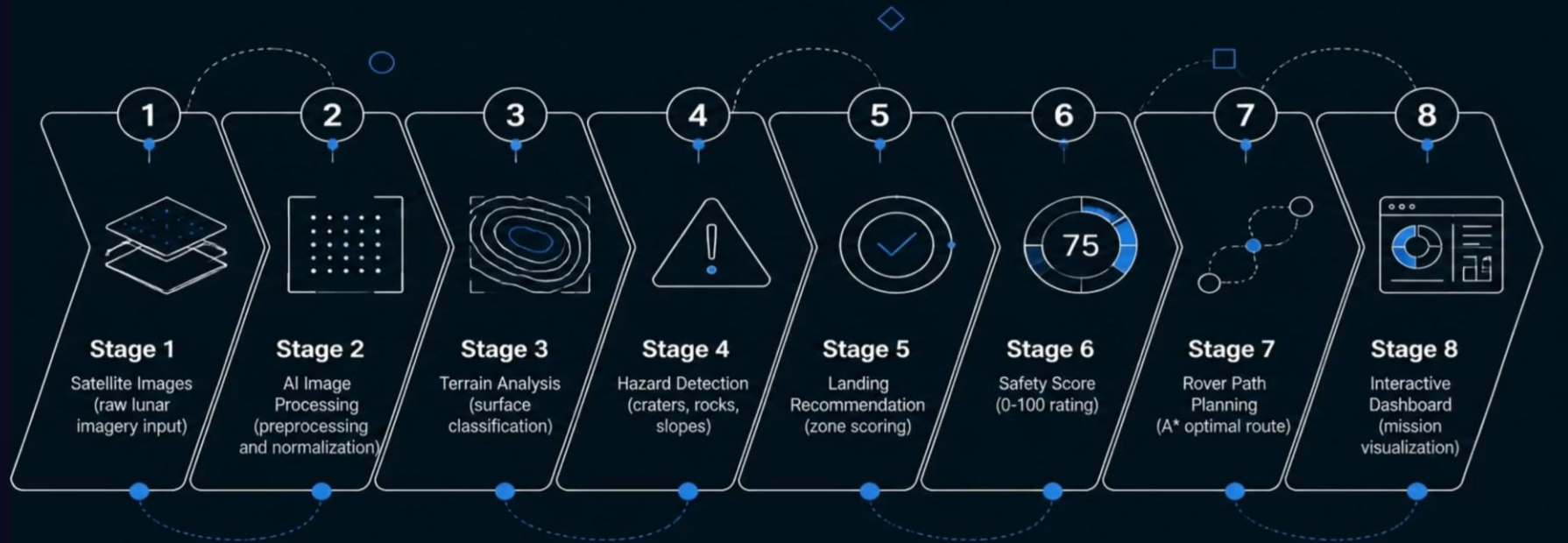
Generate hazard maps and recommend the safest landing area



Rover Path Planning

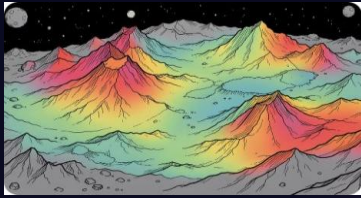
Compute the safest, most efficient traversal route for rovers

End-to-End AI Pipeline: From Image to Decision



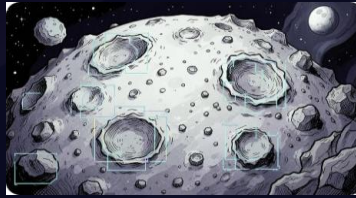
Every stage feeds the next — raw satellite imagery is transformed through AI processing, terrain analysis, and hazard detection into a ranked landing recommendation with a quantified safety score and an optimal rover path, all visualized on an interactive mission dashboard.

Five Core Modules



Terrain Analysis

Classifies surface types — flat, sloped, rough — using elevation and texture data



Hazard Detection

YOLO-based detection of craters, boulders, and dangerous terrain features



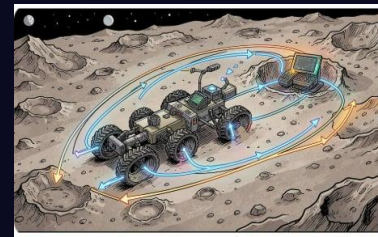
Landing Recommendation

Scores and ranks candidate zones, recommending the safest with explainability



Dashboard Visualization

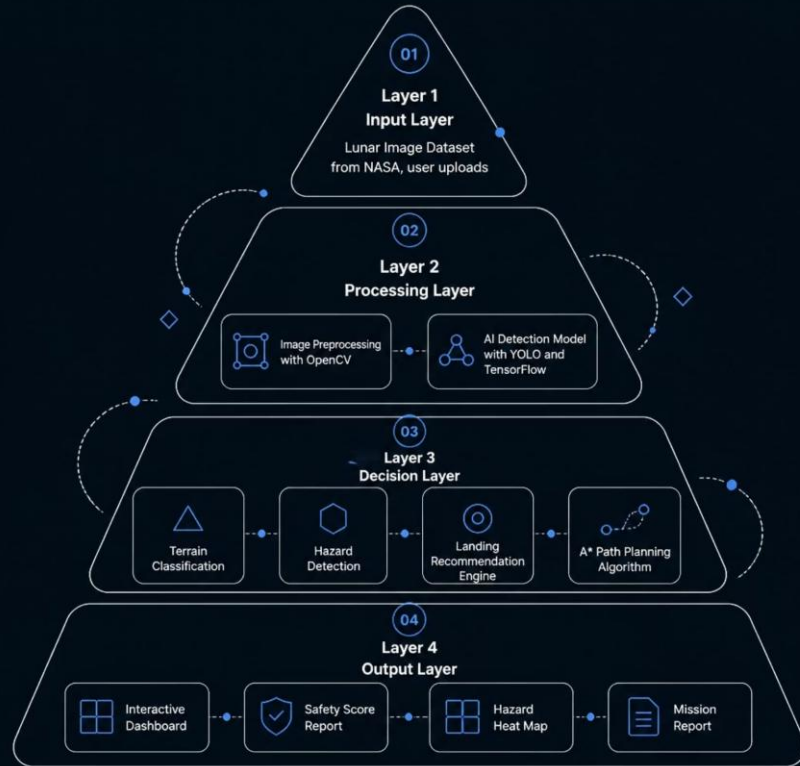
Real-time interactive mission dashboard with hazard maps, scores, and reports



Rover Navigation

A* algorithm computes obstacle-free, energy-efficient traversal paths

System Architecture



Architecture Overview

The system follows a clean four-layer architecture — from raw data ingestion through AI processing and decision-making to an interactive output layer — ensuring modularity, scalability, and real-time performance.

- **Input Layer:** NASA lunar datasets + user-uploaded imagery
- **Processing Layer:** OpenCV preprocessing, YOLO detection, TensorFlow classification
- **Decision Layer:** Terrain scoring, hazard mapping, A* path planning
- **Output Layer:** Dashboard, safety reports, heat maps, mission summaries

Technologies Powering Lunar AI



Frontend

HTML · CSS · JavaScript —
responsive mission dashboard UI



Backend

Python · Flask — lightweight
REST API for model serving



AI / ML

OpenCV · YOLO · Computer
vision ,



Algorithm

A* Path Planning — optimal, obstacle-aware
rover routing



Dataset

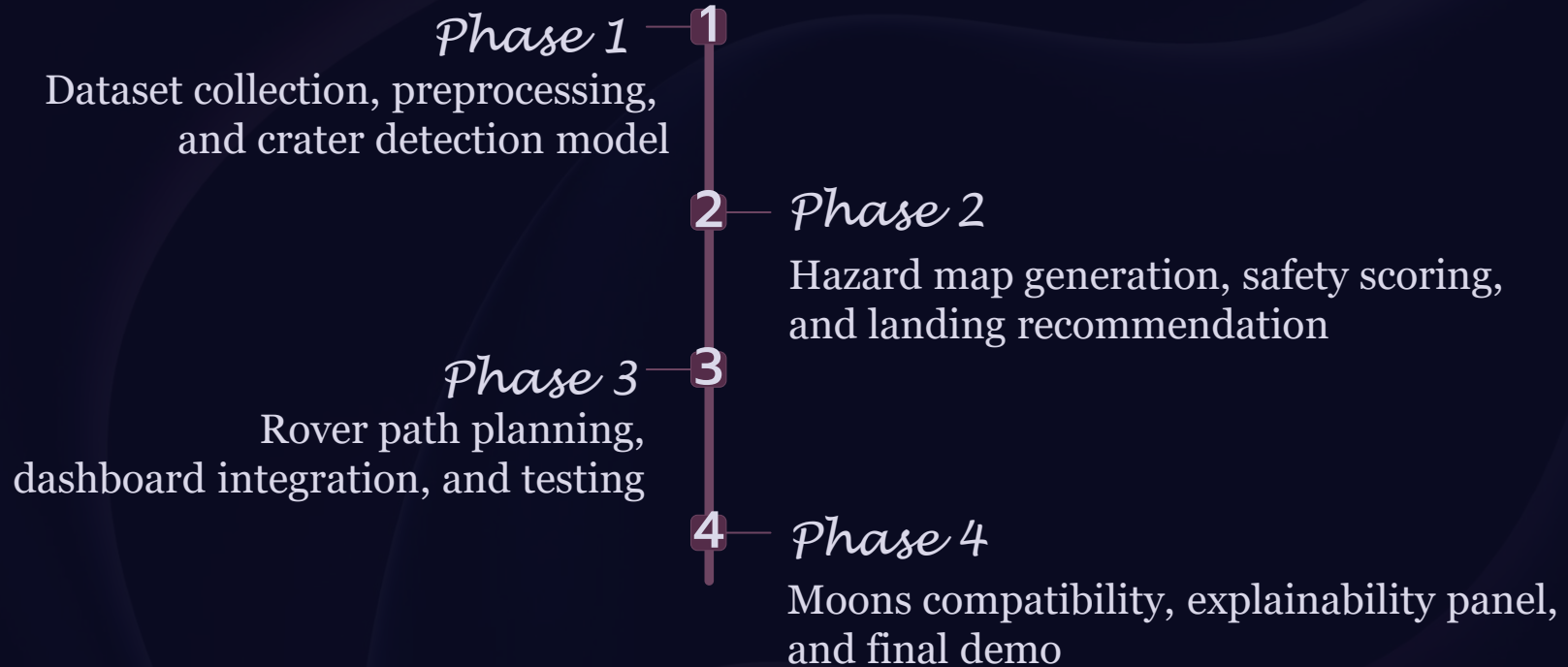
NASA Lunar Reconnaissance Orbiter Camera
(LROC) imagery database

Feature Set & Development Roadmap

 Safe Landing Detection	 Hazard & Crater Detection	 Rock & Terrain Analysis
 Safety Score & Heat Map	 Rover Route Planning	 Interactive Dashboard
	 Mission Report Generation	

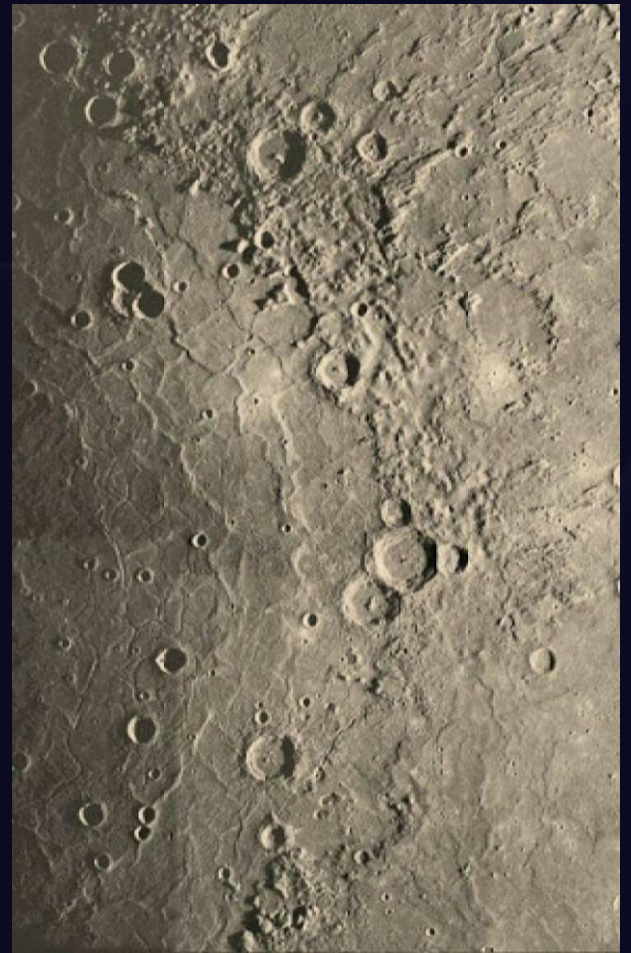
From lunar dataset to full mission dashboard — Lunar AI delivers a complete, production-ready pipeline for autonomous space exploration.

Development Roadmap



Working Methodology

Our AI pipeline transforms raw lunar imagery into actionable mission intelligence — in nine precise steps.



From Image to Intelligence



Each stage is powered by specialized deep learning models — from convolutional neural networks for terrain segmentation to reinforcement learning for optimal path planning. The entire pipeline executes in seconds, delivering mission-ready outputs to operators in real time.

Dashboard UI — Mission Intelligence at a Glance

A unified command interface presenting all critical outputs for rapid mission decisions.



Crater Detection

AI-mapped craters with size and depth classification



Rock Hazard Map

Rock clusters flagged with risk severity scores



Safe Landing Zones

Ranked zones with safety scores and coordinates



Rover Path

Optimized safe route from landing to target



Mission Statistics

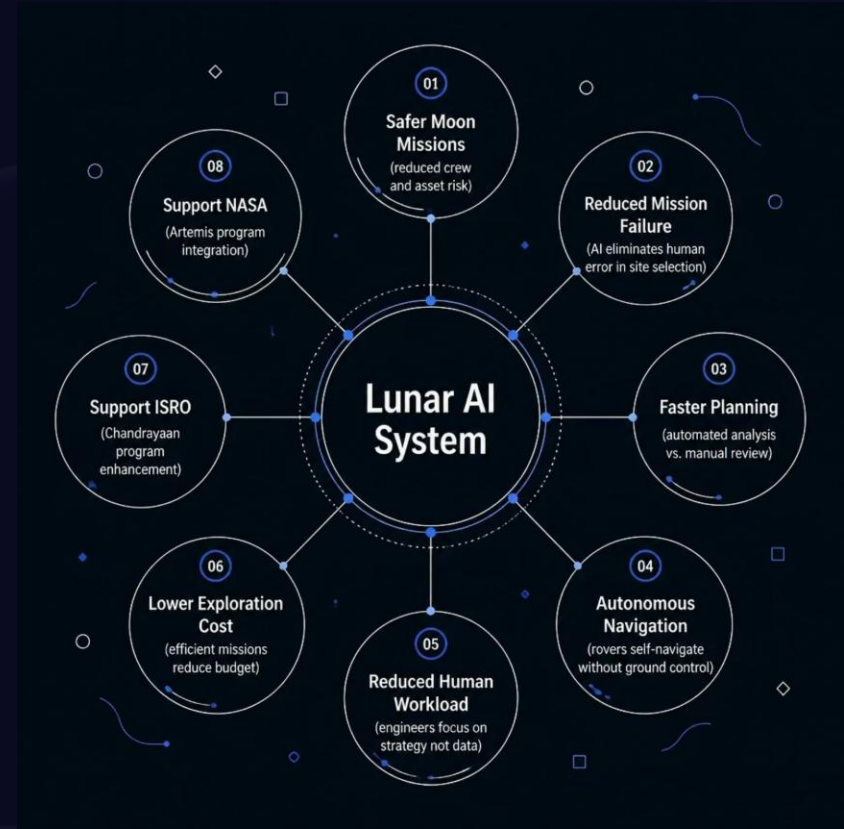
Area coverage, hazard density, and terrain metrics



Expected Impact

This system directly addresses the most critical challenges in lunar exploration — from reducing mission failure rates to enabling fully autonomous operations. By automating hazard detection and landing site selection, it dramatically shortens mission planning cycles and lowers operational costs for agencies like ISRO and NASA.

- ❏ A single successful autonomous landing can save **hundreds of millions of dollars** in mission costs and years of scientific opportunity.



Future Scope

Built to evolve — from lunar landing assistance to a full interplanetary exploration platform.



Real-Time Satellite Link

Live terrain feeds from orbiters for dynamic mission updates



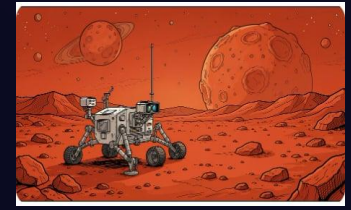
3D Terrain Mapping

Full volumetric reconstruction for precision landing analysis



Moon Digital Twin

Simulate missions in a virtual lunar replica before launch



Mars & Multi-Rover

Extend to Mars terrain with coordinated multi-rover AI missions

Technical Feasibility

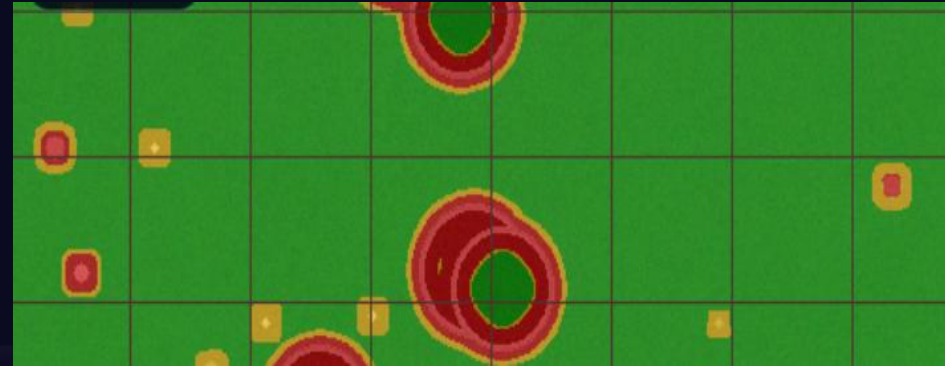
This project is grounded in proven, accessible technology — no speculative components. Every layer of the stack relies on publicly available resources, open-source frameworks, and cloud-native architecture that can be deployed within days.

Public Datasets

NASA , ISRO Chandrayaan imagery freely available

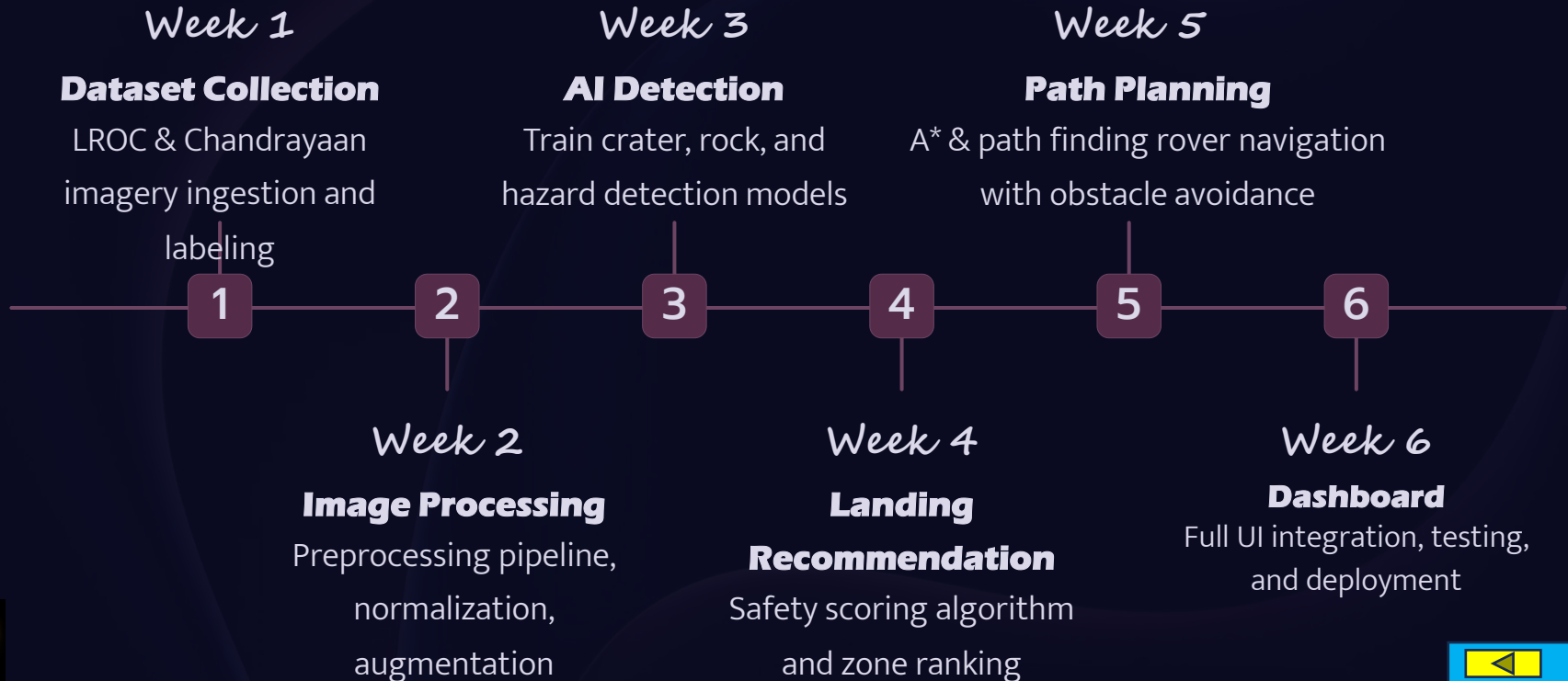
Pre-Trained Models

YOLO, OpenCV, FLASK, PIL and segmentation models ready to fine-tune



Implementation Timeline

A focused 6-week sprint from data to deployed dashboard.



Conclusion

Transforming Lunar Exploration with AI

- ❖ Our solution combines Artificial Intelligence, Computer Vision, and Smart Path Planning to make lunar landing and rover navigation safer, faster, and more reliable.
- ❖ By automating hazard detection and landing site selection, we can reduce mission risks, save valuable time, and improve exploration efficiency.
- ❖ Designed using open-source technologies and real NASA/ISRO datasets, the project is practical, scalable, and ready for future enhancements.
- ❖ Beyond the Moon, the same architecture can support Mars missions and other planetary exploration, making it a future-ready solution.

Final Thought

"The future of space exploration belongs to intelligent systems. With Lunar AI, we're not just helping machines land safely—we're enabling humanity to explore farther, smarter, and with greater confidence."

Why Our Solution Will Win



Truly Innovative

End-to-end AI for lunar missions — not just detection, but decision-making



Real-World Ready

Built on proven tech with direct application to ISRO & NASA programs



Infinitely Scalable

From Moon to Mars — the architecture extends to any planetary body



Complete Package

Modern UI, professional architecture, high feasibility, & strong visual impact

| This system doesn't just analyze the Moon —
it **protects the missions** that will one day walk on it.



THANK YOU!

"Together, let's make the next giant leap in space exploration safer with AI."

Any Questions?